

BigEasyDriver v1.6a

www.schmalzhaus.com/BigEasyDriver

An easy to use bipolar stepper motor driver
Use 4 wire, 6 wire or 8 wire stepper motors
From 0mA/phase to over 2A/phase
Defaults to 5V for Vcc (logic supply), settable to 3.3V
Supply 8V to 35V DC power input on JP1 or JP7
Do not connect or disconnect motor
while BigEasyDriver is powered

DEFAULT OPTIONS
Short JP10, or JP6 pins
to GND or Vcc to override

SLEEP = Vcc (awake)
MS1 = Vcc (1/16 microstep)
MS2 = Vcc (1/16 microstep)
ENABLE = GND (enabled)
RESET = Vcc (not reset)
MS3 = Vcc (1/16 microstep)

You only need to connect M+
GND, STEP, DIR and the motor
outputs. All other I/O is set to
default to 1/16th microstep
mode, and connections are
not required for basic
operation.

DIR is level sensitive
A rising edge on STEP
causes a step
Both take 0V to Vcc

Bi-polar Stepper Motor Outputs
Coil A of motor across
COILA+ and COILA-
Coil B of motor across
COILB+ and COILB-

Power Input JP1, JP7
8V to 35V DC

Change List:
v1.0 BPS Original version
v1.1 BPS Added gull-ups, re-routed
v1.2 5/2/11 BPS Silk screen corrections, sense resistors now .11 ohms
SparkFun design rules applied - minor layout tweaks, R13 now 2.49K
v1.3 SF 1/10/12
-updated footprint for driver IC (11)
-ran all traces directly away from IC
-moved vias and re-traced as necessary on VCC and TOP layers
v1.4 SF 4/17/2013 Fixed current adjustment pot silk so directions are correct
v1.5 SF 4/27/2013
Updated to standard 0603 footprints, some rerouting, schematic nets changed
Changed license from CC by 3.0 to CC by-sa 3.0
v1.6 BPS 5/28/2014
Removed large hole in solder mask on bottom
Changed R13 to 8.2K to improve range of current adjust pot
Added current limiting resistors to STEP and DIR

.100" Headers
D.N.I

JP10
4 SLEEP
3 RESET
2 MS3
1 MS2

JP6
2 MS1
1 ENABLE

JP14
4 DIR_IN
3 STEP_IN
2 GND
1 VCC

JP3
4 COILA+
3 COILA-
2 COILB-
1 COILB+

3.5mm Screw Terminals
D.N.I

JP11
4 SLEEP
3 RESET
2 MS3
1 MS2

JP13
2 MS1
1 ENABLE

JP12
4 DIR_IN
3 STEP_IN
2 GND
1 VCC

JP5
4 COILA+
3 COILA-
2 COILB-
1 COILB+

B340A-13-F
D1

V+

C14
47uF

GND

C1
.1uF

GND

Both C14 and C1 must
Be rated for >=50V

IN

OUT

ADJ

SPARKFUN 09_V_REG_317SOT89

R6
390

GND

3/5V Normally Open -> Vcc=5.0V
3/5V Closed -> Vcc = 3.3V

UNK3/5V0

R7
330

GND

GND

APWR Normally Shorted
Cut to use your own
Vcc source from JP14 VCC
APWR0

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R9
1K

GND

C2
1uF

GND

PWR_LED0
YELLOW

GND

GND

GND

GND

GND

GND

GND

GND

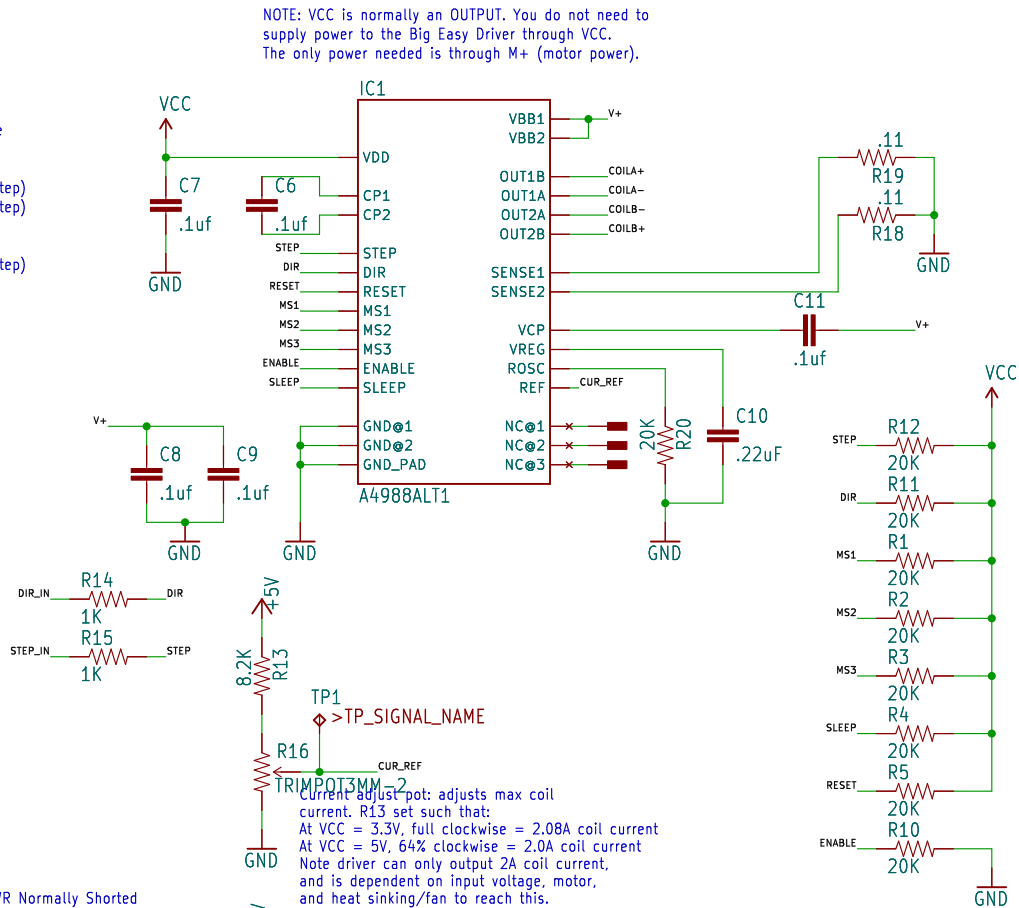
GND

GND

GND

GND

GND



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TITLE: \${PROJECTNAME}

SFE

Design by: Brian Schmalz
Revised by Patrick Alberts
Produced by SparkFun Electronics

REV:
v16a

Date: \${CURRENT_DATE}

Sheet: \${#}/\${##}

Size: User Date: Rev:
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